Sk. Md. Tashfin Sami

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Education

Bachelor of Science in Computer Science and Engineering *Jan 2020 – Dec 2025*
Patuakhali Science and Technology University, Patuakhali, Bangladesh

- CGPA: 3.25 / 4.00
- Relevant Coursework: Data Structures and Algorithms, Database Management Systems, Software Engineering, Operating System, Computer Networks, Statistics, Machine Learning

Higher Secondary Certificate in Science *Jul 2017 – May 2019*
Khulna Collectorate Public School and College, Khulna, Bangladesh

- GPA: 5.00 / 5.00

Secondary School Certificate in Science *Jan 2015 – Mar 2017*
Khulna Zilla School, Khulna, Bangladesh

- GPA: 4.77 / 5.00

Technical Skills

- **Programming Languages:** C, C++, Java, Python, HTML, CSS, JavaScript
- **Databases:** MySQL
- **Problem Solving:** 250+ problems solved on *Codeforces* , *LeetCode* and *Virtual Judge*
- **AI Integration:** OpenAI API
- **Tools and Platforms:** Git, GitHub, Postman, Linux
- **Simulators:** Cisco Packet Tracer

Projects

Blood Donation Management System | *Java, Java Swing, MySQL, JDBC* *GitHub*

- Built a desktop application using OOP to structure the system into modular classes for donors, inventory and records management.
- Implemented full CRUD operations for donor management with search and filter functionality by blood group and location.
- Integrated MySQL database via JDBC for persistent storage and designed a user friendly GUI using Java Swing.

Medical Imaging Report Generation Pipeline | *Python, Qwen3, OpenAI API* *GitHub*

- Built a pipeline that converts mammogram images into structured diagnostic reports using a large language model (Qwen3 via OpenAI Chat Completions API).
- Implemented function calling to enforce structured output fields to enforce consistent report format and reduce hallucinations.
- Applied deterministic inference techniques including temperature scaling, nucleus sampling controls, and fixed seed configuration to ensure reproducible and clinically consistent report generation.

Research & Publications

Diffusion Augmented SkinVit-Net for Skin Cancer Diagnosis | *Computer Vision, Generative AI* *IEEE*

- Published at the 28th IEEE International Conference on Computer and Information Technology (ICCIT), Cox's Bazar, Bangladesh, December 2025.
- Proposed SkinVit-Net, a lightweight MobileNetV3 Transformer hybrid classifier that achieved **93.52% accuracy** on the HAM10000 skin lesion dataset, outperforming InceptionV3 and Xception baselines.
- Addressed class imbalance in medical imaging by fine tuning a Stable Diffusion 1.5 model with LoRA to generate high quality synthetic dermoscopic images for underrepresented lesion classes.
- Evaluated synthetic image quality using LPIPS, SSIM, and PSNR metrics; model trained on Google Colab T4 GPU demonstrating feasibility in resource constrained environments.

Awards & Scholarships

Ankur Undergraduate Research Scholarship *2025*
Ankur International, Portland, USA

Reference

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